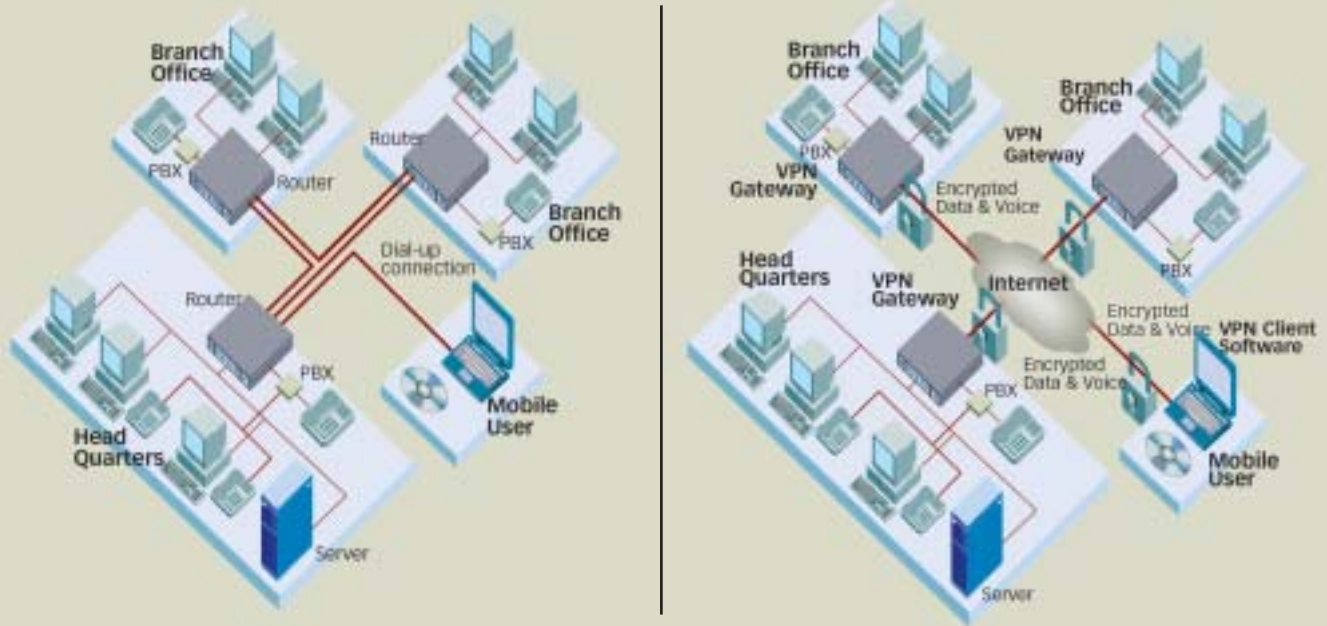


VPN Vs Leased Lines

INFOGRAPHIC: Solomon Lewis



A visual comparison of VPN and leased lines reveals the simplicity of the former

tunnels. The number of tunnels supported by a gateway depends on the hardware and the speed of the connection. The gateway performs functions such as initiation, authentication and managing the tunnels after the connection is set up. The gateway then routes the data to the corresponding node in its network. It also encrypts the data before sending it, and decrypts the data that is received. The VPN client provides remote access to a VPN gateway, and performs similar functions, such as authentication, encryption and decryption. Apart from these, the client also authenticates the user who wants to access the corporate network. The security of a VPN is managed at three points—the client, the target network and the gateway.

Info The VPN services and equipment market in India is expected to grow to Rs 1,276 crores by 2005, up from Rs 168 crore in 2001. **Source: IDC**

The client is not a replacement for anti-virus software or security updates. The client should be virus free and have all security updates before connecting. The target network should also have a limited access set for remote users. This way, even if an unauthorised user gains access to the network, the possible risk and damage is minimised. It is also important to remember that data is unencrypted after it passes the gateway, and can flow freely in the network. Thus, a secure gateway is also crucial to maintain security in a VPN.

The architecture of any network is like a building—like the floors of a building, a

network is made up of layers. The lowest layer is the physical medium, such as coaxial cable, twisted pair cables, etc. The top-most layer is the layer where applications such as browsers work. The intermediate layers, from top to bottom, are the presentation layer, sessions layer, transport layer, network layer and data link layer. When data is transmitted, it flows from the application layer to the intermediate layers of the sender's computer, travels over the physical medium, and then moves to the application layer in the destination system. Each protocol operates at a specific layer in the network; for example, IP operates in the Network layer. Similarly, the PPTP operates in Layer 2 of the network.

PPTP, IP... huh?!

PPTP (Point to Point Tunneling Protocol), L2F (Layer 2 Forwarding), L2TP (Layer 2 Tunneling Protocol) and IPSec (IP Security) are VPN protocols that are used to transmit packets. The main function of these protocols is to encrypt data and tunnel it through the Internet.

PPTP is an extended version of the PPP (dial-up protocol) that provides support for VPN. It was developed by a group of companies led by Microsoft. Similarly, Cisco developed the L2F protocol for secure private dial-up networks over the Internet. The L2TP protocol is an IETF (Internet Engi-

neering Task Force) standard that combines the best of the PPTP and L2F. It can be used wherever PPTP or L2F is being used today. The IPSec protocol was also developed by the IETF, and operates in the IP layer of the network. IPSec offers two modes of encryption—Transport and Tunnel. Each packet has a data portion and a header portion, with details such as destination and source address. In the transport mode, only the data portion of a packet is encrypted, whereas in the tunnel mode, both the header and the data portions are encrypted. Though criticised for its complexity, this is the most secure protocol currently available.

MPLS VPN

There are two types of VPNs. IP VPNs operate at layer 2 of the network, and are based on a connection-oriented model—a connection has to be established between the sender and the receiver before the data can be transmitted, as in a telephone conversation. In the connectionless model there is no need to establish a connection for data transfer—it is similar to sending an e-mail or an SMS. A VPN based on the connectionless model is more scalable than a connection-oriented one, and is recommended for larger networks. This is because large networks deal with thousands of tunnels, which may lead to lower throughput, or variable bandwidth if the connection-oriented model is being used.

MPLS (Multi-protocol Label Switching) networks operate at layer 3 of a net-